

Industrial Equipment Monitoring Platform

Requirements Specification

Version: 1.0 — Draft baseline

Project owner: Jerry

Release: Portfolio MVP

Delivery estimate: Six weeks at 8–12 hours per week, subject to milestone review

The targets in this specification are proposed acceptance criteria, not measured results or a claim of standards certification.

1. Product purpose

Build a platform that helps maintenance operators monitor equipment temperature, identify disconnected devices, investigate historical readings, and acknowledge alerts.

The platform must demonstrate an end-to-end workflow across embedded firmware, a Linux gateway, backend services, and a web dashboard.

The primary demonstration is:

A sensor detects excessive temperature and an operator sees an alert. During a ten-minute internet interruption, the gateway stores readings locally. After connectivity returns, it uploads the backlog without creating duplicate records.

2. Scope and assumptions

The MVP supports one organisation with multiple sites. Users receive access to specific sites.

The physical demonstration uses one ESP32, one temperature sensor, and one Raspberry Pi gateway. A simulator provides additional devices for testing.

Monitoring is advisory. Automatic equipment control, emergency shutdown, and safety certification are outside the project scope.

Included in MVP	Deferred
Device and site registry	Customer self-service and billing
Real temperature readings and simulated devices	Additional industrial protocols
Persistent gateway buffering and replay	ESP32 offline flash buffering
Historical telemetry and connection status	Predictive maintenance and AI
Temperature alerts and acknowledgement	Email, SMS, and escalation schedules
Authentication and site permissions	Enterprise SSO
Docker deployment and CI	Kubernetes and high availability
Operational logs and backup/restore procedure	Fleet OTA firmware updates

AWS deployment is an extension after the MVP acceptance tests pass.

3. Users and permissions

Role	Permissions
Administrator	Manage users, site access, sites, gateways, devices, credentials, and alert rules
Operator	View assigned sites, inspect telemetry, and acknowledge alerts
Viewer	Read dashboards and history for assigned sites
Gateway service	Submit telemetry only for devices assigned to that gateway

Permissions must be enforced by the backend. Hiding controls in the dashboard is insufficient.

4. Proposed implementation baseline

Component	Technology	Responsibility
Sensor firmware	ESP32, ESP-IDF, C/C++	Acquire temperature and publish telemetry
Local messaging	Mosquitto, MQTT	Transport readings from ESP32 to gateway
Linux gateway	Raspberry Pi, C++17, SQLite	Validate, persist, batch, and retry readings
Backend	FastAPI, PostgreSQL	Registry, ingestion, authentication, queries, and alerts
Background worker	Python, shared backend codebase	Connection checks and retention jobs
Dashboard	React, TypeScript	Device status, charts, and alert workflow
Delivery	Docker Compose, GitHub Actions	Repeatable deployment and automated verification

The gateway forwards telemetry to the backend using HTTPS batches. The backend remains one application with a supporting worker.

5. Functional requirements

“Must” identifies a release requirement. Each requirement must link to an implementation issue and verification evidence.

ID	Requirement	Acceptance criteria
FR-01	Administrators must register sites, gateways, and devices.	A device has a unique ID, name, site, assigned gateway, and enabled state. Duplicate IDs are rejected.
FR-02	Users must authenticate before accessing operational data.	Invalid, expired, or missing credentials are rejected. Disabled accounts cannot access protected resources.
FR-03	Access must be restricted by role and site.	An operator assigned to Site A cannot read or change Site B resources by modifying API identifiers.
FR-04	Firmware must publish real temperature readings.	The ESP32 publishes a reading every five seconds by default. A disconnected sensor produces an explicit error status rather than a fabricated temperature.
FR-05	Every reading must have a stable event identity.	Retries retain the same device ID, boot ID, and sequence number. A new boot receives a new boot ID.

ID	Requirement	Acceptance criteria
FR-06	The gateway must persist readings before forwarding them.	Readings committed to SQLite survive a gateway process restart and remain queued until confirmed by the backend.
FR-07	The gateway must recover from temporary backend failure.	Connection failures, timeouts, rate limiting, and server errors trigger bounded exponential backoff with jitter. Recovery requires no manual restart.
FR-08	Ingestion must be idempotent.	Sending the same event ten times creates one telemetry record. Reusing its identity with different content is rejected and logged.
FR-09	Ingestion must validate each batch item.	Unknown devices, wrong gateway ownership, unsupported schema versions, invalid units, and malformed values receive explicit rejection reasons. Valid items in a mixed batch can still succeed.
FR-10	The dashboard must show current equipment status.	Each device shows its latest temperature, reading age, connection state, and active alert count. Missing data is displayed as unavailable.
FR-11	Users must be able to inspect historical readings.	Users can select a device and time range. Charts use measurement time where trustworthy, show units, and do not imply continuous data across gaps.
FR-12	The system must detect stale connections.	With a 30-second heartbeat and a 90-second timeout, a connection becomes offline within 120 seconds of its last heartbeat. Historical replay does not mark it online.
FR-13	Administrators must configure a high-temperature rule.	A rule specifies its threshold, number of consecutive readings, and recovery threshold. Invalid configurations are rejected.
FR-14	The system must manage alert episodes.	Three consecutive valid readings above the configured threshold create one active alert. Further high readings do not create duplicate active alerts.
FR-15	Operators must be able to acknowledge alerts.	Acknowledgement records the user and time. It does not resolve the temperature condition. Viewers cannot acknowledge.
FR-16	Important administrative actions must be auditable.	User-access changes, credential revocation, rule changes, and acknowledgements record actor, action, target, and UTC time.

6. Telemetry contract and processing rules

Field	Requirement
schema_version	Required supported payload version
device_id	Required registered device identifier
boot_id	Required identifier unique to the device boot
sequence_number	Monotonically increasing number within that boot
measured_at	UTC measurement time when the device clock is synchronised; otherwise null
device_uptime_ms	Elapsed time since device boot
gateway_received_at	UTC time assigned when the gateway receives the reading

Field	Requirement
backend_received_at	UTC time assigned by the backend
metric	temperature for the MVP
value	Finite numeric value within the selected sensor's documented range
unit	celsius
quality	Explicit reading and clock-quality information

The database must enforce uniqueness on (device_id, boot_id, sequence_number).

The backend derives gateway identity from its credential and verifies device ownership. It must not trust a caller-supplied site identifier as proof of access.

Processing rules:

- Backend acceptance is returned only after the database transaction commits.
- Batch responses identify accepted, duplicate, and rejected events individually.
- The gateway removes accepted or matching duplicate events from its queue.
- Permanently invalid events move to a bounded quarantine with a rejection reason.
- Transient failures remain queued.
- Older events may populate history but must not replace a newer current reading.
- All stored timestamps use UTC. The dashboard displays the selected timezone explicitly.
- Maximum batch size is 500 readings; larger batches are rejected.

MQTT QoS 1 provides at-least-once delivery, so duplicates remain possible and require application-level handling. A broker acknowledgement does not prove that the gateway has committed a reading to SQLite. [1]

For this MVP, the recovery guarantee begins when a reading is committed to gateway storage. Loss before that point must be documented.

7. Alert behaviour

The initial demonstration rule uses configurable test values:

- Trigger: temperature above 30°C for three consecutive valid readings.
- Recovery: temperature below 28°C for three consecutive valid readings.
- A gap longer than 15 seconds resets a pending consecutive-reading count.
- Invalid readings do not count toward triggering or recovery.
- Readings more than 30 seconds old when received by the backend are stored as history but do not change live temperature alerts.
- Duplicate and out-of-order readings do not advance the live alert evaluation.
- Missing telemetry produces a connection-status change; it does not prove that a temperature alert has recovered.

An alert has separate condition and acknowledgement information. It may be active and acknowledged, or resolved and unacknowledged.

8. Reliability performance and operational requirements

These are release-test targets. Record the hardware, software versions, dataset, network conditions, and actual results.

ID	Requirement	Acceptance target
NFR-01	Normal ingest capacity	Sustain 100 simulated devices at one reading per second for 30 minutes, with all valid events eventually stored once.
NFR-02	Dashboard freshness	At the normal test load, p95 time from gateway receipt to dashboard visibility is at most five seconds.
NFR-03	Query response time	With one million seeded readings and ten concurrent clients, p95 latency is below 500 ms for device lists and one-device, 24-hour aggregated history queries.
NFR-04	Offline storage	Buffer at least 100,000 readings on the declared gateway hardware.
NFR-05	Recovery	After a ten-minute backend outage at 100 readings per second, drain the 60,000-reading backlog within ten minutes while new readings continue.
NFR-06	Restart durability	Restart the gateway during an outage; every event committed before restart eventually reaches the backend exactly once as a stored record.
NFR-07	Storage exhaustion	When the configured queue limit is reached, preserve queued events, reject new intake visibly, and increment a loss counter. Never silently overwrite data.
NFR-08	Bounded queries	History endpoints enforce time-range and result limits. Charts return no more than 2,000 aggregated points per series.
NFR-09	Retention	Retain raw telemetry for seven days by default. A scheduled job removes expired records without deleting registry data.
NFR-10	Recovery procedure	Create a daily database backup and demonstrate restoration into a clean environment within 30 minutes. The nominal backup recovery point is up to 24 hours old.
NFR-11	Observability	Expose queue depth, oldest queued-event age, ingest rate, rejected events, retries, and service health.
NFR-12	Usability	Core dashboard actions work with a keyboard. Status uses text or icons as well as colour. Empty, loading, and error states are explicit.

9. Security requirements

Security testing must address relevant OWASP API risks, particularly object-level and function-level authorisation. This is a verification reference, not a certification claim. [2]

ID	Requirement	Verification
SEC-01	Use HTTPS for deployed web/API traffic and authenticated TLS for device MQTT connections.	Untrusted certificates and unauthenticated broker clients are rejected.
SEC-02	Give each gateway a separate revocable credential.	A revoked credential cannot submit a new batch.
SEC-03	Restrict device MQTT permissions.	A device cannot publish on another device's telemetry topic.

ID	Requirement	Verification
SEC-04	Store passwords using a maintained password-hashing implementation.	Plaintext passwords are absent from the database and logs.
SEC-05	Keep secrets outside source control.	Repository scanning passes and example configuration contains placeholders.
SEC-06	Limit login attempts and request sizes.	Repeated failed logins are throttled; oversized requests are rejected.
SEC-07	Test authorisation at every protected resource boundary.	Automated tests cover cross-site access, viewer writes, and gateway device impersonation.
SEC-08	Protect operational data from accidental disclosure.	Application errors omit stack traces and secrets; logs exclude passwords and tokens.
SEC-09	Review dependencies before release.	No unresolved critical or high findings remain without a documented assessment and mitigation.

10. Mandatory end to end acceptance scenarios

Test	Scenario	Expected result
AT-01	Publish a real sensor reading.	The correct device temperature appears on the dashboard with its timestamp and unit.
AT-02	Publish three consecutive excessive readings, acknowledge, then publish three recovery readings.	One alert opens, retains its acknowledgement details, and resolves after recovery.
AT-03	Disconnect the backend for ten minutes while local MQTT remains available.	The gateway queue grows; recovery uploads all committed readings without duplicate records.
AT-04	Restart the gateway while the backend is unavailable.	The durable queue survives and resumes forwarding after recovery.
AT-05	Commit a batch at the backend but interrupt its response.	The gateway retries safely and database counts remain correct.
AT-06	Submit mixed valid, invalid, and duplicate readings.	Each item receives the correct outcome; invalid events cannot block valid delivery indefinitely.
AT-07	Attempt cross-site access and unauthorised alert acknowledgement.	Access is denied and data remains unchanged.
AT-08	Stop the device while replaying old telemetry.	The device becomes offline; replay does not make it appear live.
AT-09	Reach the gateway's configured storage limit.	Existing data remains intact and rejected intake is measurable.
AT-10	Deploy from a clean checkout and restore a backup.	The documented procedure produces a working system with restored records.

11. Delivery milestones

Milestone	Target	Exit criteria
M1 — Contracts and vertical slice	Week 1	Requirements, telemetry schema, database migrations, and simulator-to-API ingestion work.
M2 — Physical demonstration	Week 2	ESP32 readings travel through the gateway and appear on the dashboard.
M3 — Reliability	Week 3	Durable buffering, retry, duplicate protection, and restart tests pass.
M4 — Operational workflow	Week 4	Authentication, site permissions, connection status, and alert workflow pass acceptance tests.
M5 — Deployment and verification	Week 5	CI, deployed environment, security checks, and backup/restore procedure work.
M6 — Release evidence	Week 6	Performance results, known limitations, documentation, and a three-minute demonstration are published.

Authentication must be enabled before exposing operational endpoints publicly, even if the full permissions milestone comes later.

If delivery exceeds the available time, defer extensions before weakening the reliability or authorisation criteria.

12. Definition of done and handover

A requirement is complete when:

- Its implementation issue links to the requirement ID.
- Relevant automated or documented hardware tests pass.
- Error behaviour is verified.
- API and operational documentation reflect the implementation.
- No blocking defect remains.

The MVP release must include:

- Source repository with reproducible setup instructions.
- Architecture overview and database model.
- Versioned API and telemetry contracts.
- CI results and acceptance-test evidence.
- Benchmark report containing measured results and test conditions.
- Deployment, backup, restore, and troubleshooting instructions.
- Three architectural decision records covering transport, buffering, and duplicate protection.
- A short demo showing normal operation, failure, and recovery.
- A known-limitations list.

Changes to the baseline must record the affected requirement IDs, reason, acceptance-test changes, and schedule impact. CV and portfolio claims must use verified results from the released implementation.

References

[1] OASIS. MQTT Version 3.1.1. OASIS Standard, 29 October 2014. Section 4.3.2, QoS 1 at least once delivery.

<https://docs.oasis-open.org/mqtt/mqtt/v3.1.1/os/mqtt-v3.1.1-os.html>

[2] OWASP. OWASP Top 10 API Security Risks 2023. API1 Broken Object Level Authorization and API5 Broken Function Level Authorization.

<https://api-security.owasp.org/editions/2023/en/0x11-t10/>